EXPERIMENT 3

AIM

To develop a Python-based knowledge-driven object identification system that analyzes images using a pretrained image recognition model and applies a predefined knowledge base and rule-based reasoning to identify objects, their categories, properties, and functions.

ALGORITHM

1. Start the program.
2. Import the required libraries for image processing and object recognition.
3. Create a predefined knowledge base containing object names, categories, properties, and functions.
4. Accept an input image in JPG, JPEG, or PNG format.
5. Read and display the input image using Python.
6. Load a pretrained image recognition model.
7. Analyze the image and identify the object with its confidence score.
8. Check whether the confidence score meets the predefined threshold.
9. If the confidence is low or the object is not in the knowledge base, display an appropriate message.
10. Otherwise, retrieve the object's information from the knowledge base.
11. Apply rule-based reasoning to determine the object's category and function.
12. Display the object name, category, properties, and description.
13. Test the system using different images and evaluate its correctness.
14. Stop the program.

CODE:

```
# Knowledge-Based Object Identification System


# Knowledge Base

knowledge_base = {

   "laptop": {

      "category": "Electronic Device",

      "properties": "Screen and keyboard",

      "function": "Computing and learning"
```

```
},
"book": {
    "category": "Educational Object",
    "properties": "Pages and cover",
    "function": "Reading and studying"
},
"chair": {
    "category": "Furniture",
    "properties": "Seat, backrest and legs",
    "function": "Sitting"
},
"bottle": {
    "category": "Container",
    "properties": "Hollow body and narrow opening",
    "function": "Storing liquids"
},
"mobile phone": {
    "category": "Electronic Device",
    "properties": "Touchscreen and portable design",
    "function": "Communication"
},
"table": {
    "category": "Furniture",
    "properties": "Flat surface and supporting legs",
    "function": "Placing objects"
```

```
    }
}


# Input object
print("Available objects:", ", ".join(knowledge_base.keys()))
obj = input("Enter the object name: ").lower().strip()


# Rule-based reasoning
if obj in knowledge_base:
    info = knowledge_base[obj]
    print("\nObject Identified:", obj)
    print("Category:", info["category"])
    print("Properties:", info["properties"])
    print("Function:", info["function"])
else:
    print("Object could not be identified.")
```

OUTPUT:

